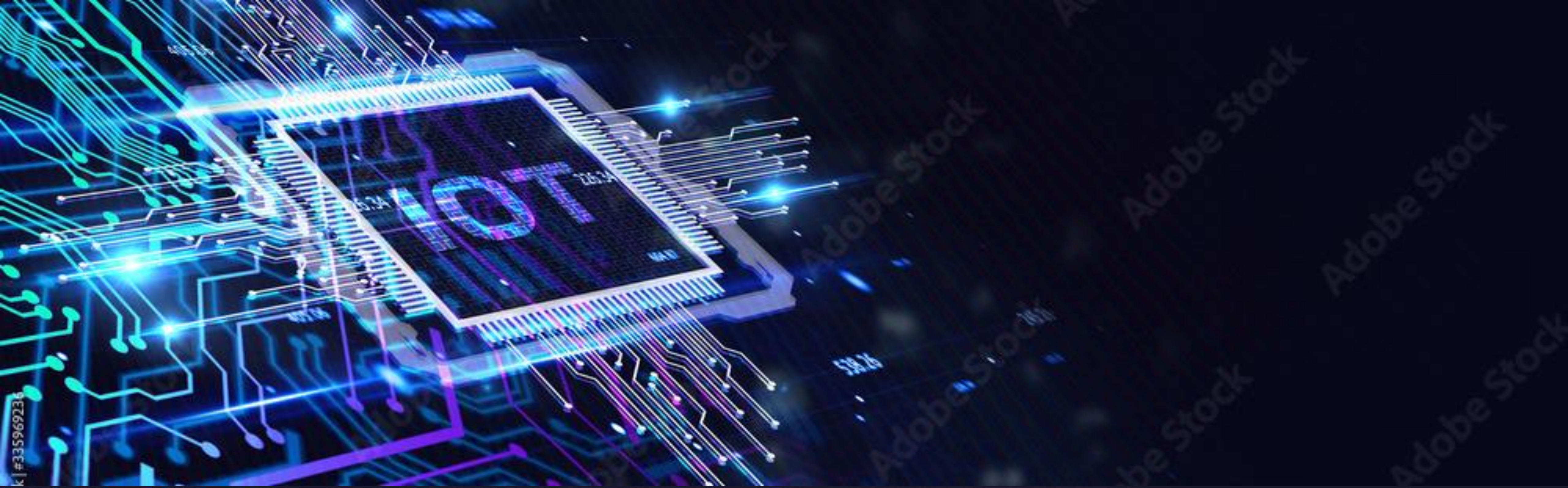
Internet of Things (IoT)

Topics

- Introduction
- IOT Definition
- Applications



IoT -

- Internet of Things (IoT) is the network of physical objects – things – that are embedded with sensors, software and other technologies which communicates and exchange data over the internet.
- Things can be physical objects like vehicles, buildings, appliances and other devise.

Internet Of Things

Any Device

Anybody

Anywhere

Any Business

Any Network

Anytime

Applications of IoT

- Agriculture
- Tracking (Vehicle, Pets, Human, Asset)
- Energy section
- Safety and security
- Defense
- Education
- Water management
- Health care
- Smart City

Characteristics of IoT

1. Connectivity

   Any one, any where, any time connectivity should be guaranteed.

2. Intelligence and identity

   Each IoT device have a unique ID (IP address), which can be used to read sensor data.

3. Scalability

   Should be capable of handling massive expansion.

4. Dynamic and self adapting

   IoT devices should dynamically adapt changing context or surroundings.

Characteristics of IoT ....

5. Architecture

   The architecture should be hybrid which supports products from different manufactures.

6. Safety

   The data security of the device and the network should be ensured.

Things in IoT

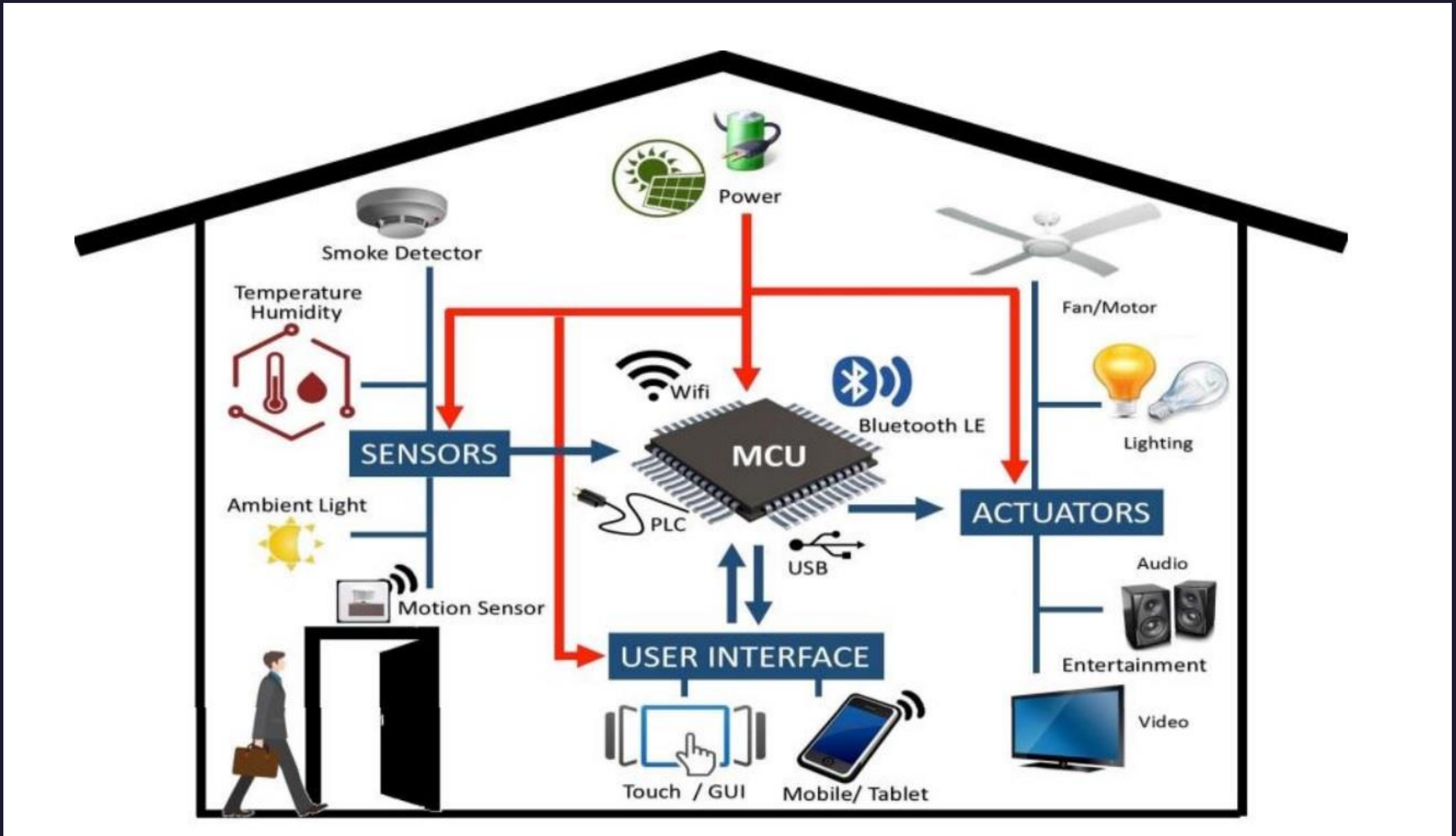
- In IoT, things refers to variety of devise.
- Things can monitor or measure data.
- Things can exchange data with other devices in the system.
- Examples:
    - Vehicles
    - Wearables
    - Home appliances

Things = Hardware + Software + Service

Things in Home Automation System

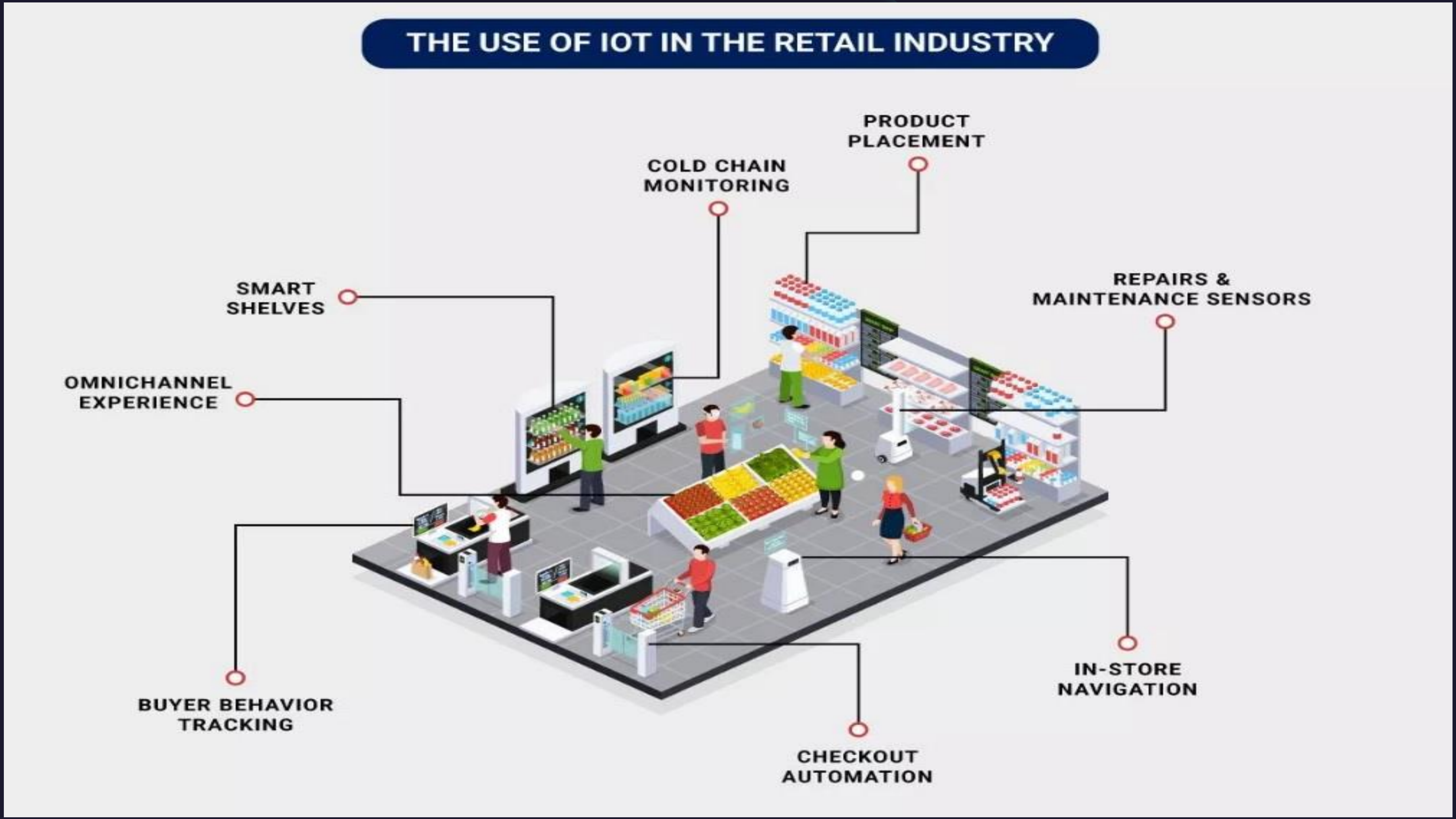
Things in Home Automation

- Lighting devices
- Air conditioner
- Fan
- Washing machine
- Refrigerator
- Security cameras
- Smart phones
- Door locking system



Name the things in a Smart Retail System

Things in a Smart Retail System



IoT Stack

- IoT stack has seven different layers.
- Each layer defines protocols for the functioning of that layer

ISO – OSI Layers

Application	7
Presentation	6
Session	5
Transport	4
Network	3
Data Link	2
Physical	1

IoT Stack ...

Layer – 1 (Physical or Sensor Layer):

- This layer handles physical components such as sensors.
- This layer is responsible for data collection.
- Example for sensors – temperature sensor, humidity sensor, pressure sensor, light sensor etc.

IoT Stack ...

Layer – 2 (Processing and control action Layer):

- This layer includes microprocessor or microcontroller.
- The data collected from the sensor is processed here.
- It uses development boards such as Arduino, ESP32, Raspberry pi etc
- It uses operating systems like Linux, RTOS, Android, IOS etc.

IoT Stack ...

Layer – 3 (Hardware Interface Layer):

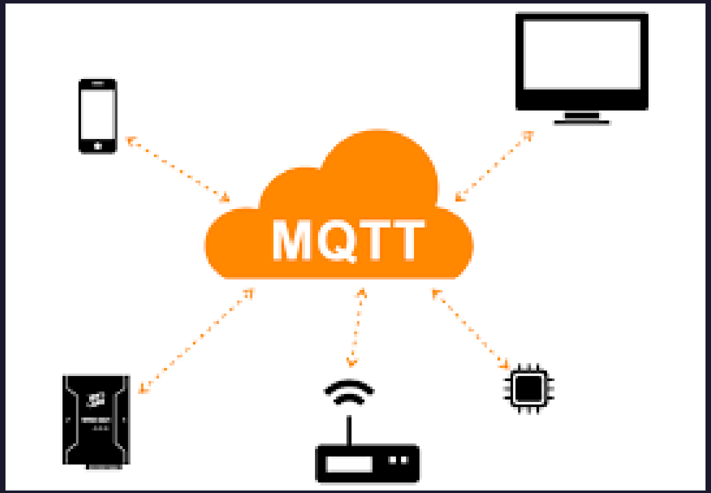
- This layer defines the communication standard used for exchanging data in an IoT network.
- It includes : USB, SPI, SCI, I^2C, RS232, CAN

IoT Stack …

Layer – 4 (RF Layer):

- RF – Radio Frequency plays a major role in communication channels.
- Protocols used for communication based on RF is listed in this layer.
- Some of the RF protocols are : Wi-Fi, NFC, RFID, Bluetooth.
- It also includes Li-Fi

IoT Stack ...



Layer – 5 (Session/Message Layer):

- This layer defines how messages are broadcasted.
- It uses protocols like : MQTT, CoAP, SSH, FTP etc.

IoT Stack ...

Layer – 6 (User experience Layer):

- This layer concerned with the end user experience.
- It includes programming languages, scripting languages and data bases.
- Example: C++, Java Script, SQL etc.

IoT Stack ...

Layer – 7 (Application Layer):

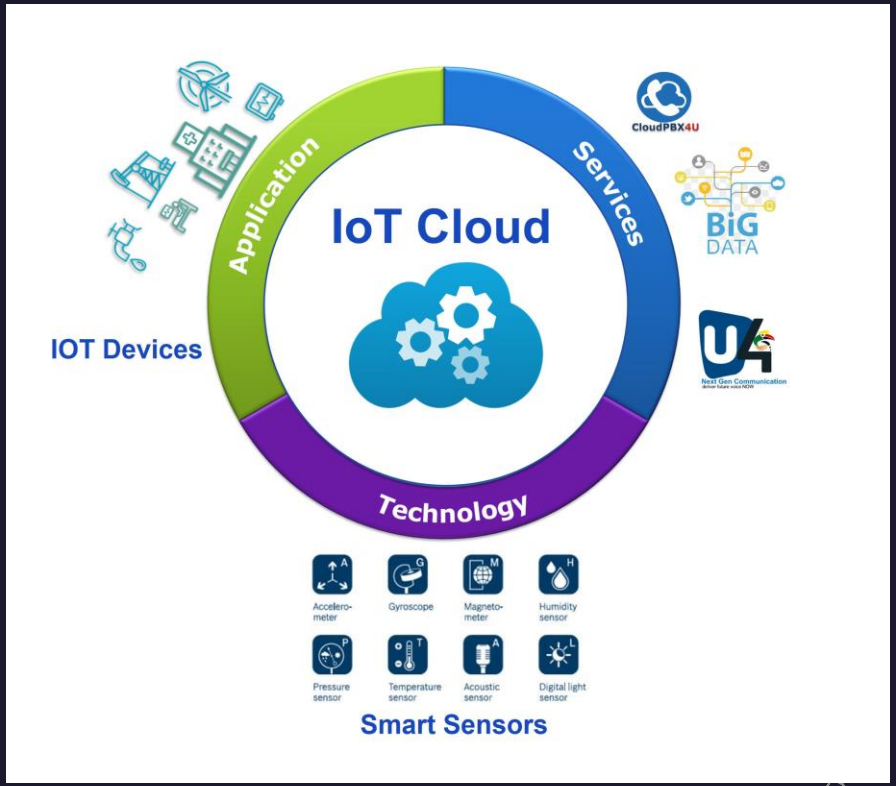
- This layer defines the applications which can be built with the support of other layers.
- Examples: smart home, smart city, smart parking, smart retail, smart agriculture, smart energy etc.

IoT Stack

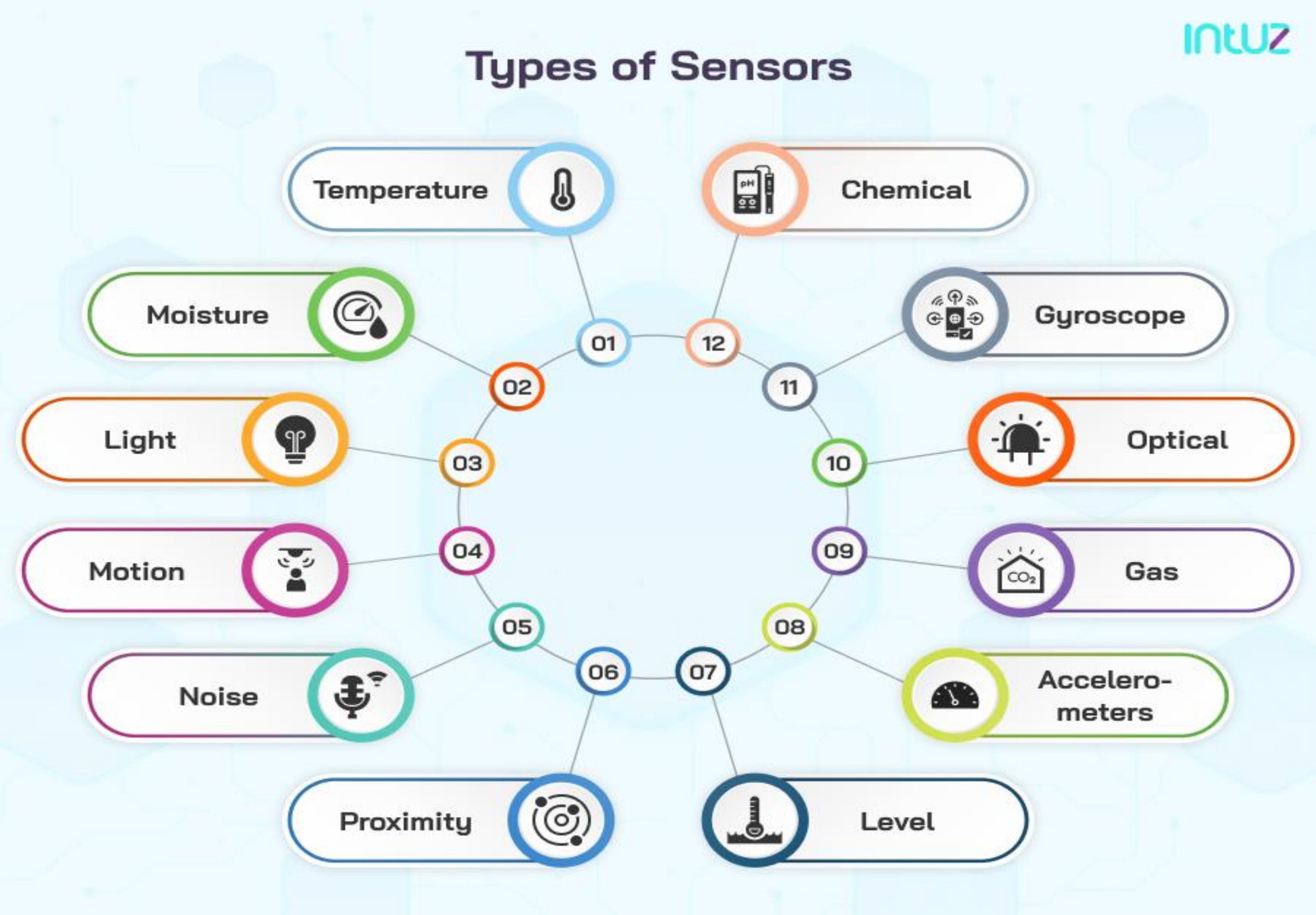
- Application Layer
- User Experience Layer
- Session/Message Layer
- RF Layer
- Hardware Interface Layer
- Processing and Control action Layer
- Physical or Sensor Layer

IoT Enabling Technologies

- IoT is a collection of many technologies or devices.
- IoT enabling technologies can be categorized into:
  1. Technologies that help to collect data.
  2. Technologies that help to analyse data.
  3. Technologies that takes control actions.
  4. Technologies that enhance security.

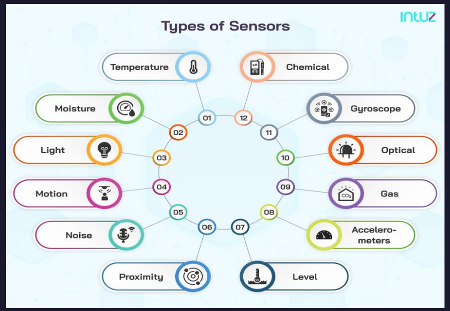


Sensors:



Types of Sensors
Temperature
01
Moisture
02
Light
03
Motion
04
Noise
05
Proximity
06
Level
07
Accelero-
meters
08
Gas
09
Optical
10
Gyroscope
11
Chemical
12

Sensors:

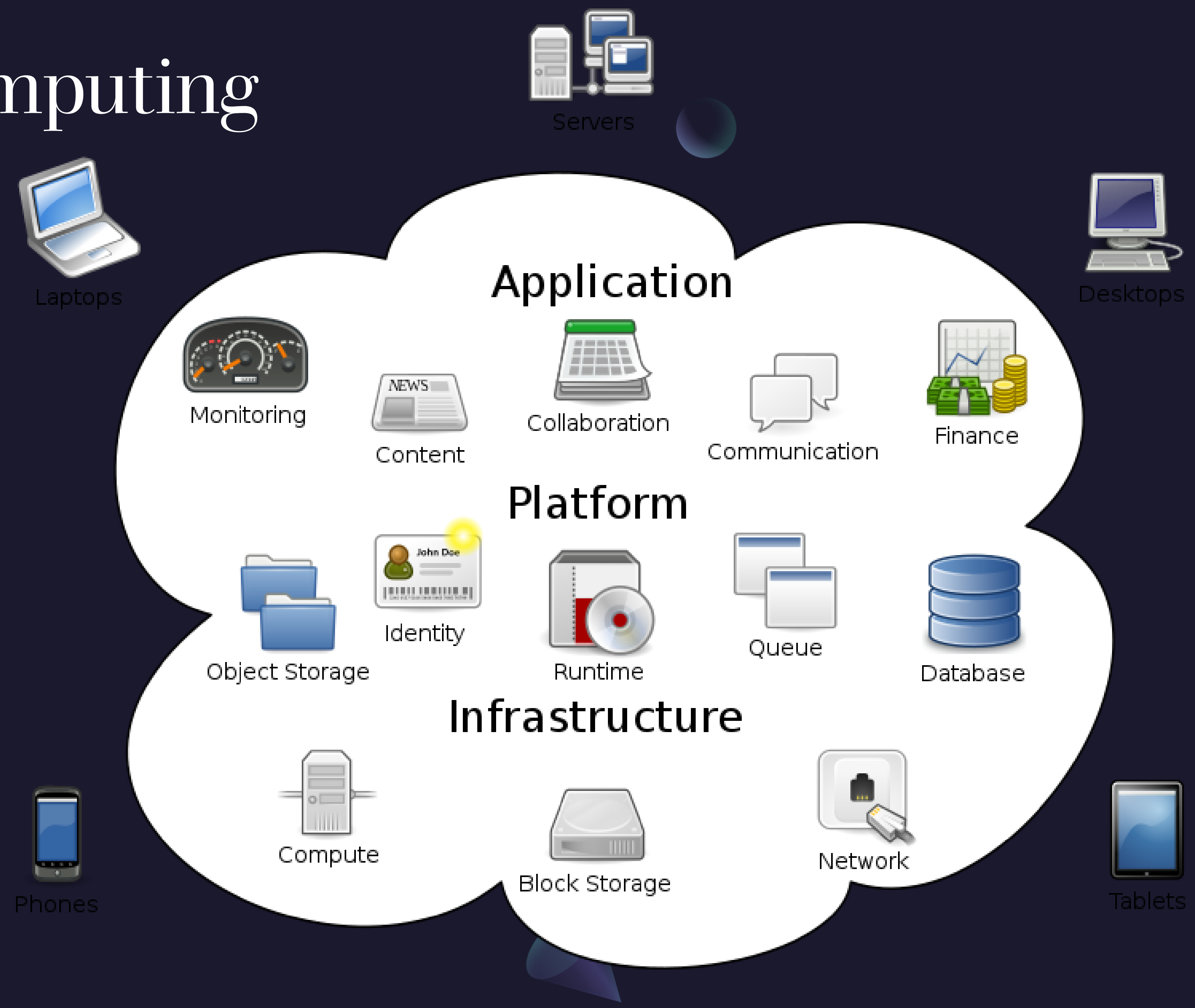


- Sensors are the heart of any IoT application.
- They sense the environment and collect data.
- Sensors are the starting point of any IoT application

Example of using Sensors as enabling technologies:

- Camera used in home automation system.
- Weather tracking system using temperature/humidity/moisture sensor.
- Vehicle monitoring system detecting speed and tyre pressure.
- Water quality monitoring using sensors that measure PH, chloride etc.
- PIR sensor used in signals for pedestrian crossing detection.

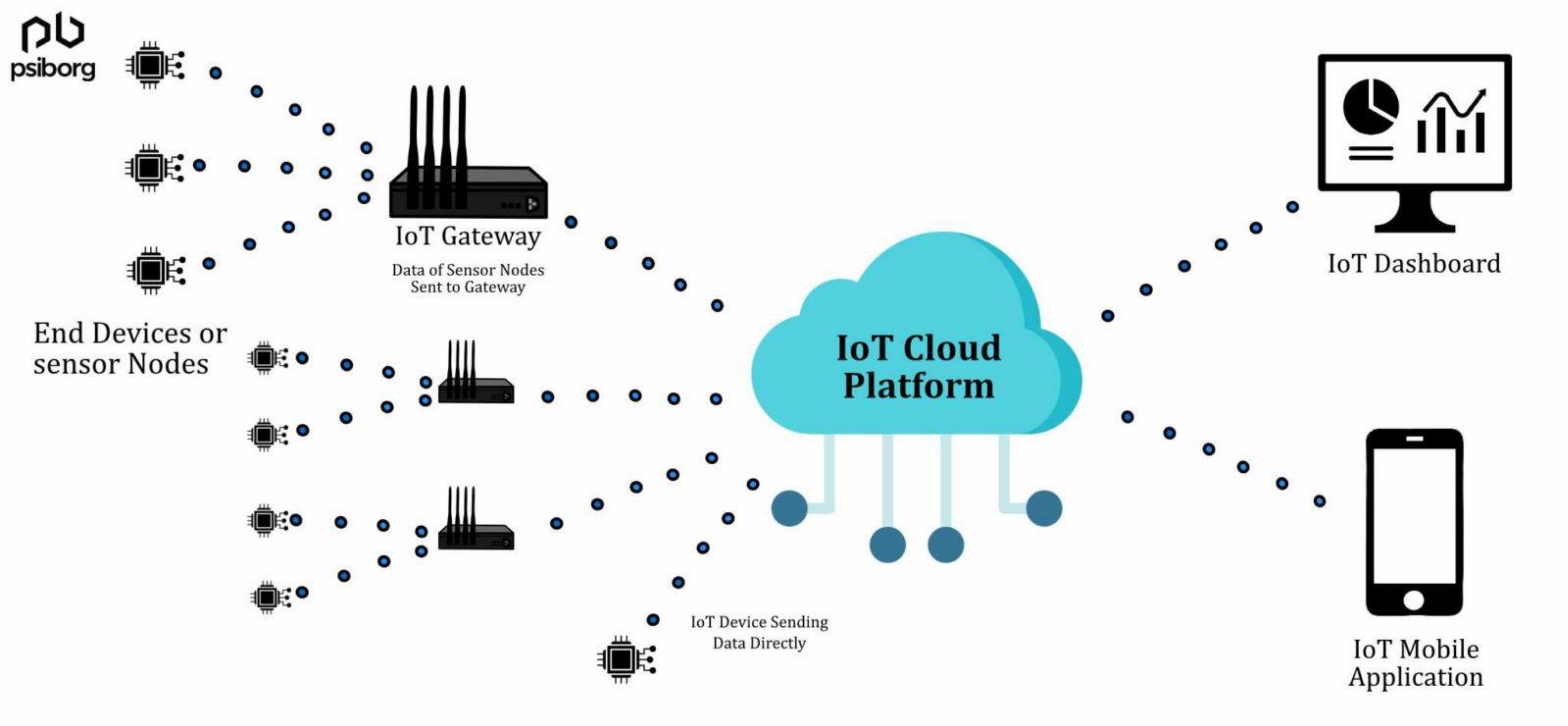
Cloud Computing



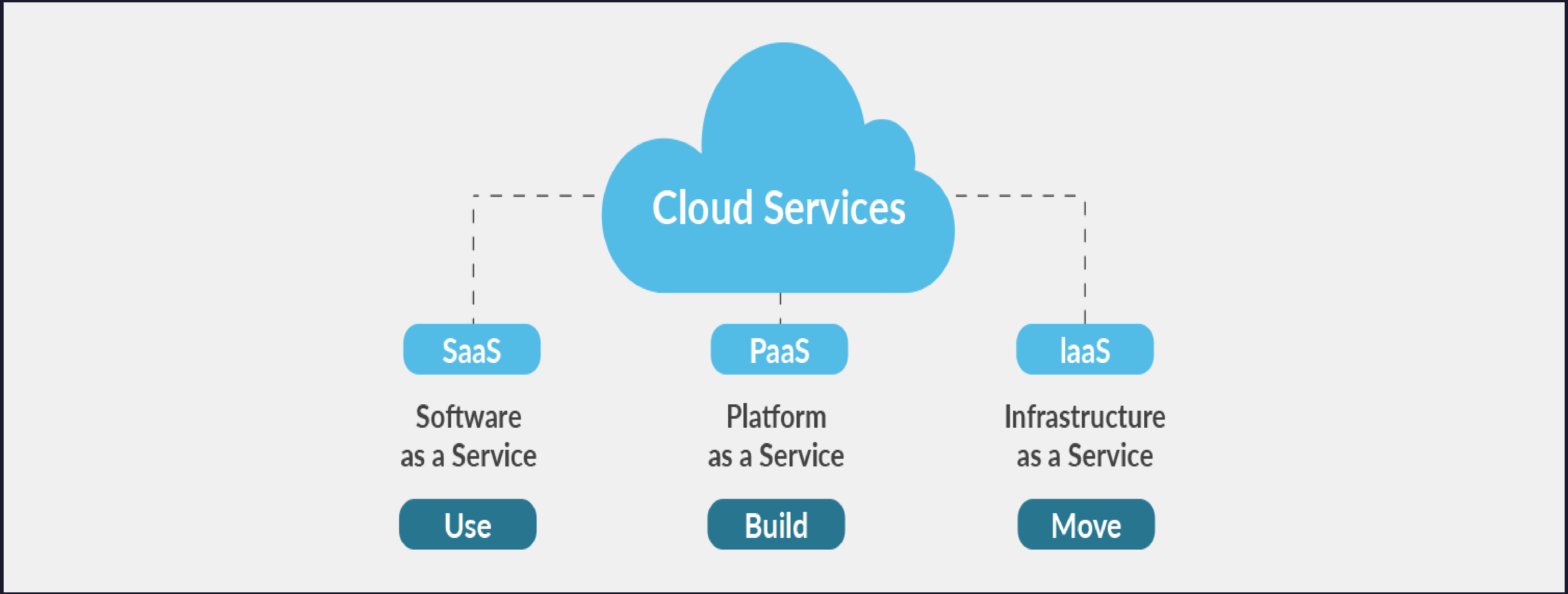
Cloud Computing

- Cloud computing is the on-demand availability of computing resources such as storage space, servers, database, software, networking, analytics etc over the internet.
- Cloud is used as data storage for IoT applications.
- Some of the popular Cloud computing providers are:
- Amazon web service(AWS), Microsoft Azure, Adafruit, ThingSpeak

Cloud Computing in IoT



Cloud Services



Cloud Services

- IaaS (Infrastructure as-a-service):
  - This cloud service provides virtual machines that can be customized by the user.
- PaaS (Platform as-a-service):
  - This cloud service provides the hardware and software tools needed to application development over the internet.
- SaaS (Software as-a-service):
  - This cloud service provides complete software applications to end user.
  - The service can be availed in subscription.

Big data analytics

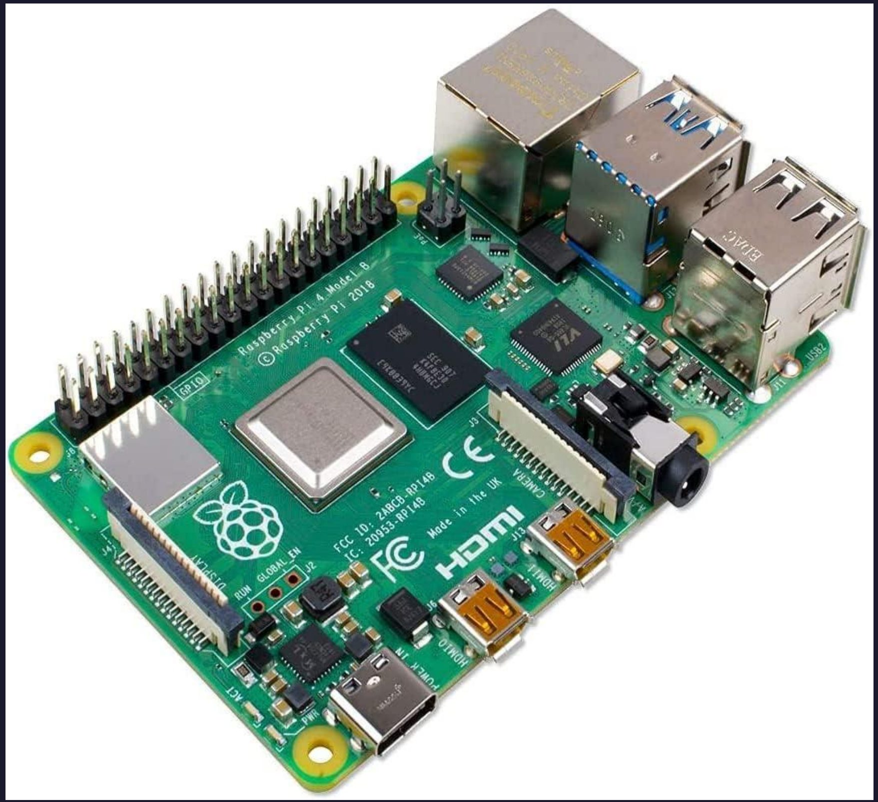
Big data analytics

- Data analytics is one of the enabling technology for IoT applications.
- Every sensor in IoT application generates data.
- The big data analytics will be based on:
- 1. Scale (Volume): Huge amount of data is generated in every second.
- 2. Complexity(Variety): Data comes from different sources in different format.
- 3. Speed(Velocity): The data is generated very rapidly and it should processed accordingly.
- 4. Accuracy(Veracity): The accuracy of the data is a problem. The data may be corrupted or incomplete.

Computing Boards

Arduino

NodeMCU



Raspberry Pi

Computing Boards...

- Embedded computing boards are required to build IoT applications.
- Most of the computing boards includes Microprocessors or Microcontrollers.
- Some of the computing boards are:

1. Raspberry Pi
2. Arduino
3. NodeMCU
4. Intel Edison

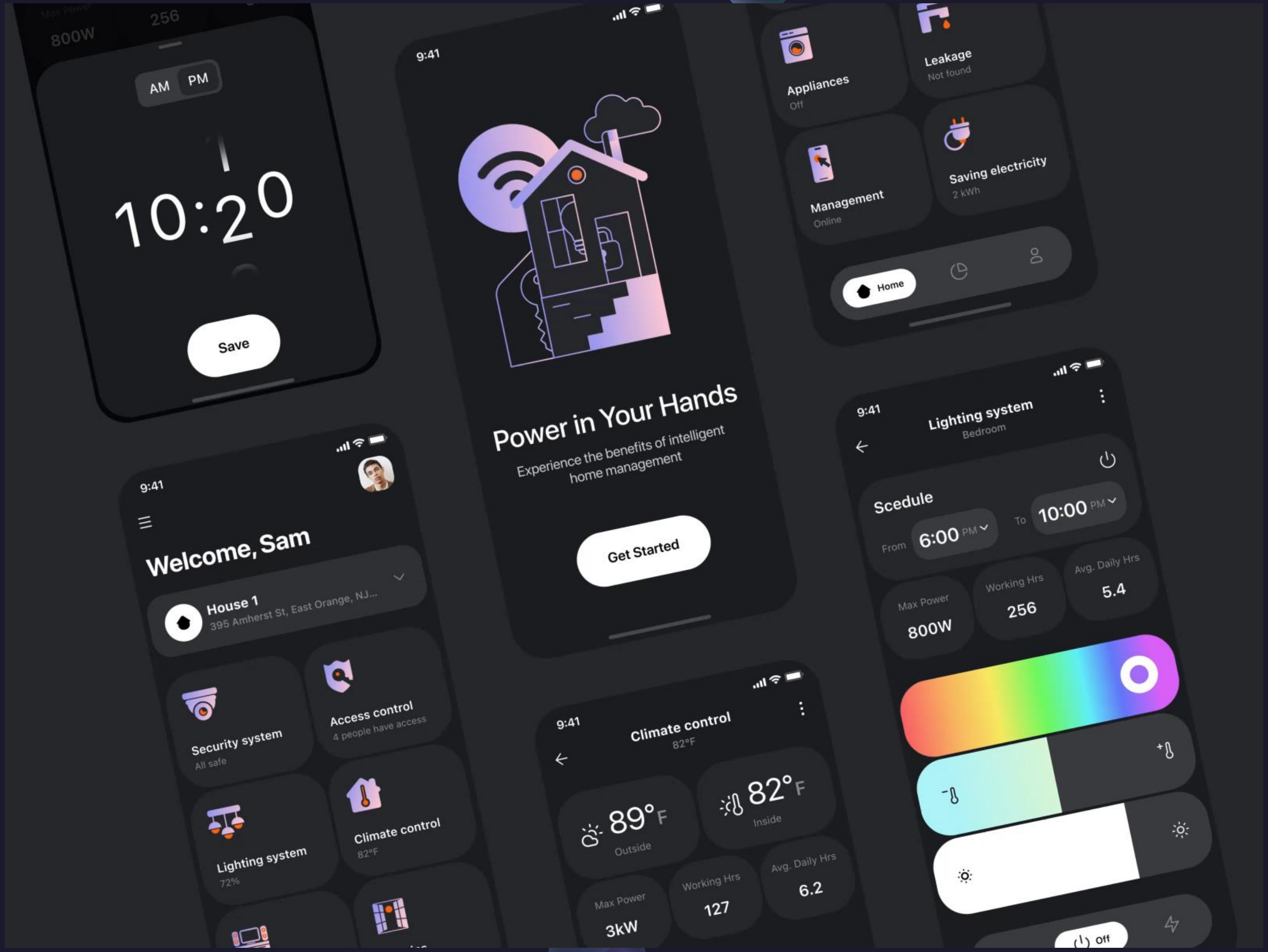
Communication Protocols

- Protocols are the rules that defines the data exchange between devices.

Communication Protocols

- Protocols defines the following:
  1. Addressing
  2. Format of message
  3. Routing
  4. Flow of control
  5. Message security
  6. Segmentation of packets
  7. Retransmission
  8. Error monitoring

User Interfaces



User Interfaces...

- User interfaces are used to monitor or configure IoT applications.
- User Interface design should be simple.
- Most IoT application user interfaces will be Mobile app or web application.

IoT Challenges

- While Building an IoT application, we face many challenges.
- It includes both technical and non technical.

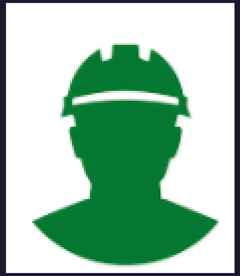
IoT Challenges...

- The following are some of the challenges:

1. Security / Personal safety
2. Privacy
3. Data extraction
4. Connectivity
5. Power requirement
6. Complexity
7. Storage

1. Security / Personal safety

DATA SECURITY



PERSONNEL SAFETY

- It is the most important challenge
- Since a number of devices are used in IoT, user data theft will happen.
- Since, many devices are connected in the loop, if one device get attacked, other devices could also be attacked.
- People's personal safety is a concern and challenge due to implants or use of sensors.

2. Privacy

I NEED PRIVACY

- We could be tracked or monitored by anyone, as we are connected 24x7 in the internet.
- Could be tracked without our permission.
- It is a challenge for us to overcome this.

3. Data extraction

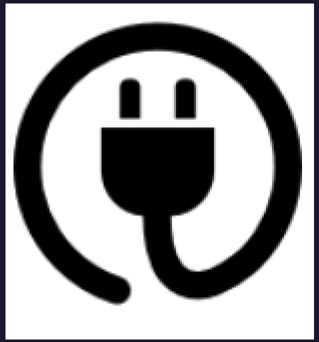
- Data extraction will be difficult in complex environment.
- Internet connectivity many not be available in all places.

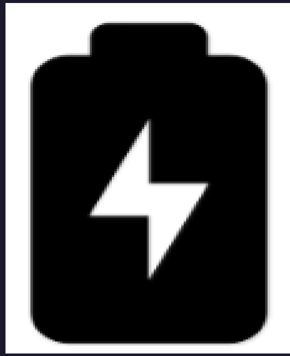
4. Connectivity

- Wired or wireless connectivity is required for IoT applications.
- Usage of the frequency / spectrum is also to be considered. (like 2.4 Ghz band)

5. Power requirement

- All the IoT devices needs power.
- Most of them are battery operated.
- Need for more power and long lasting batteries are challenges.





6. Complexity

- IoT devices need a lot of technologies to work together.
- Limited expertise in the field and rapid growth.
- Limited toolkits, software and hardware to design applications.

7. Storage

- Cloud is becoming mandatory for storing and analyzing data in IoT.
- The challenges related to storage are,

  -Which cloud to choose?

  -How much it costs?

  -What are the services it provides? etc.

LEVELS OF IoT

With the amount of complexity involved, IoT applications are classified into different levels.

- There are 5 levels.
  - Level – 1
  - Level – 2
  - Level – 3
  - Level – 4
  - Level – 5

Level - 1

- This is the simplest level.
- Have one sensor / device to sense. (could be temp sensor/pressure sensor etc.)
- Less amount of data generated.
- The data to be stored in locally.
- Data analysis to be done.
- Monitoring / control can be done through a Mobile app or Webapp.

Example:
Home automation system

Level - 2

- Single node
- **Sensing frequency is higher**
- **Huge amount** of data generated. (**Big data**)
- **Data is stored in a cloud.**
- The data analysis is done locally.
- Monitoring / control can be done through a Mobile app or Webapp.

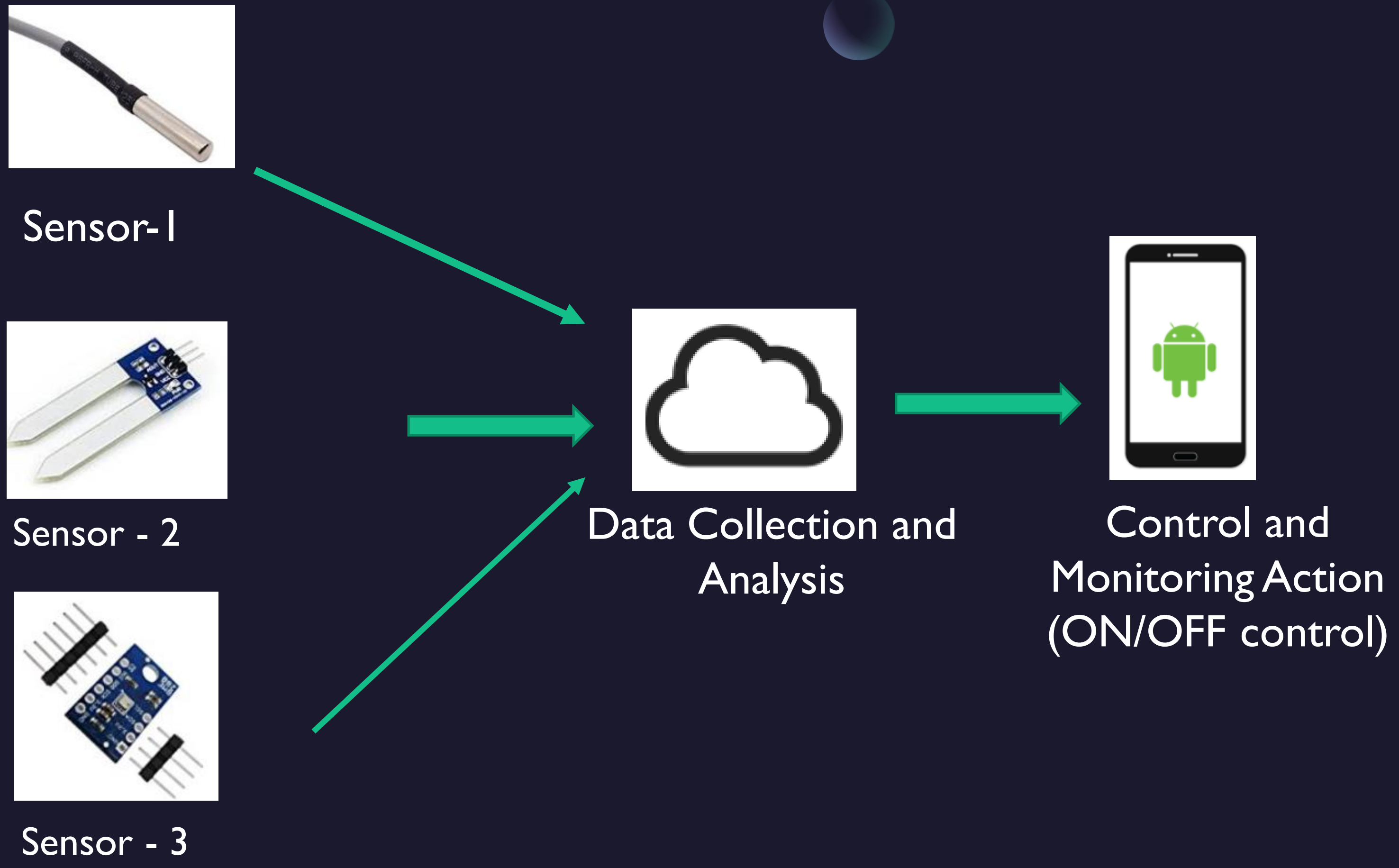
Example:
Smart Irrigation

Level - 3

- Single node.
- Sensing frequency is higher
- Huge amount of data generated. (Big data)
- Data is stored in a cloud.
- **The data analysis is done in the cloud.**
- Monitoring / control can be done through a Mobile app or Webapp.

Example:
Package tracking

Level - 4

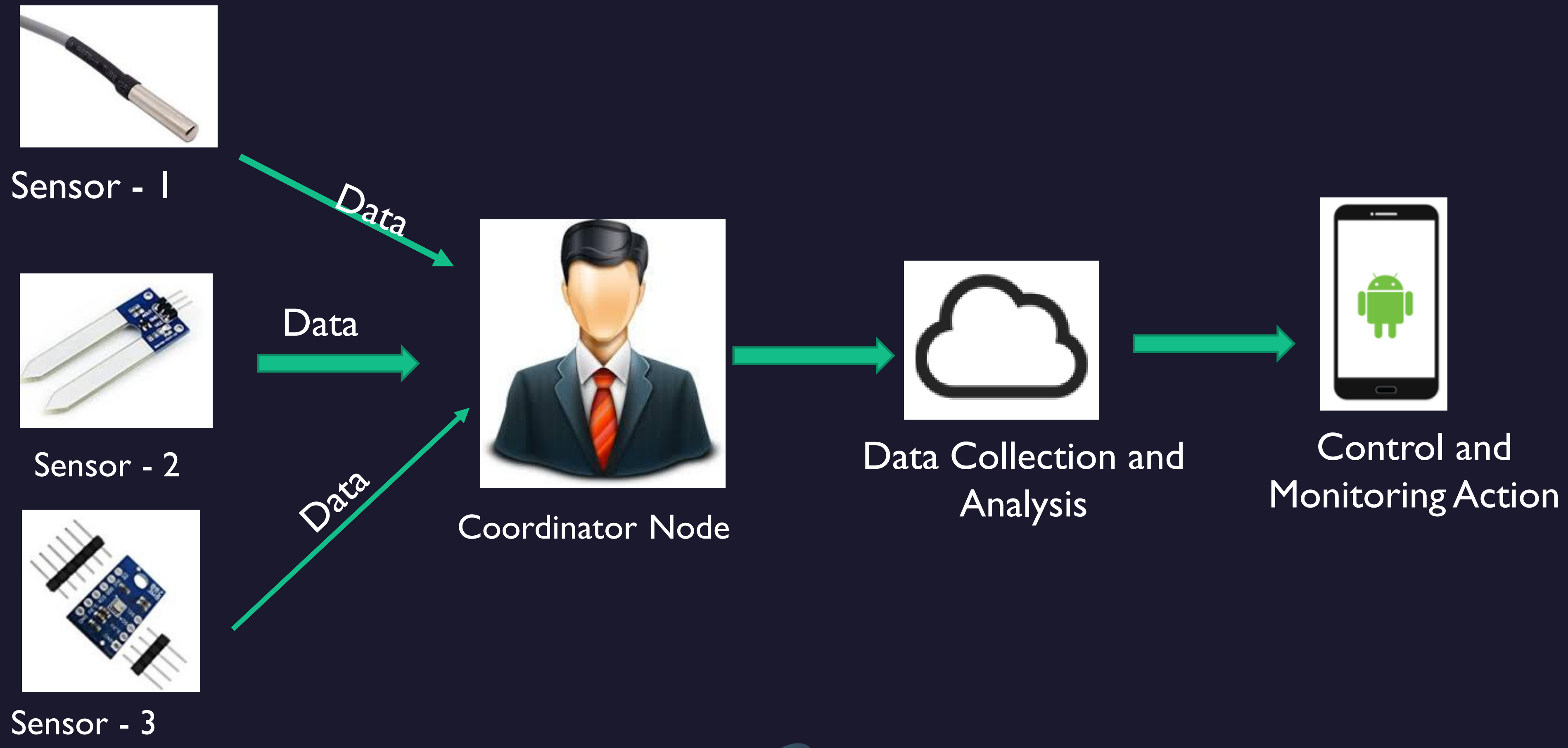

Sensor-1
Sensor - 2
Sensor - 3
Data Collection and Analysis
Control and Monitoring Action (ON/OFF control)

Level - 4

- **More than one independent nodes.**
- Sensing frequency is higher
- Huge amount of data generated. (Big data)
- Data is stored in a cloud.
- The data analysis is done in the cloud.
- Monitoring / control can be done through a Mobile app or Webapp.

Example:
Noise monitoring system

Level - 5


Sensor - 1
Data
Sensor - 2
Data
Data
Sensor - 3
Coordinator Node
Data Collection and Analysis
Control and Monitoring Action

Level - 5

- More than one independent nodes.
- Sensing frequency is higher
- Huge amount of data generated. (Big data)
- **Data is collected by coordinator node and send to cloud.**
- Data is stored and analysed in the cloud.
- More computation required.
- Monitoring / control can be done through a Mobile app or Webapp.

Example:
Forest fire detection and prevention

Thank You

Summary